Language-Guided Progressive Query Refinement With Diffusion Model for Visual Grounding on Remote Sensing Images

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***Abstract*—Remote sensing visual grounding (RSVG) is a lively research topic in the RS community. Due to the superior performance of transformers in cross-modal interaction and alignment, they have dominated the RSVG community, and numerous transformer-based methods have been developed. Although they have achieved success in various applications, their performance in visual and textual feature alignment still does not reach the level we expected. In addition, these methods place relatively limited emphasis on bounding box coordinate generation, where directly predicting coordinates through a decoder and grounding head can result in suboptimal localization accuracy. To overcome the above limitations, this article presents a new RSVG method, named language-guided progressive query refinement with diffusion model for visual grounding (LPQR-DVG). It contains a multiscale text-guided feature learning module (MTFLM) and an iterative query localization module (IQLM). MTFLM integrates text guidance into all stages of the visual feature extraction process through cross-attention layers, effectively incorporating textual information into the visual representation. In this way, text-aware visual features can be effectively obtained, which facilitates cross-modal interaction and target localization. IQLM improves cross-modal feature alignment and iterative target localization by introducing an RSVG-oriented reversed U-Net with progressive query refinement (PQR) units. Through the PQR units in IQLM, visual and text features are progressively aligned via multiple rounds of interaction, enabling the query to capture the target information referred to by the text gradually. Finally, a diffusion-based iterative training strategy is introduced to help LPQR-DVG map the random bounding box coordinates into the final predictions progressively. The encouraging results counted on the DIOR-RSVG and OPT-RSVG datasets confirm the effectiveness of LPQR-DVG. Our source codes are available at https://github.com/TangXu-Group/LPQR-DVG**

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***Index Terms*—Diffusion model, language guidance, progressive reasoning, remote sensing (RS), visual grounding.**

I. Introduction

REMOTE sensing visual grounding (RSVG) is a lively research topic in the RS community. It focuses on localizing the referred objects within RS images according to the textual descriptions [1], [2], which can be viewed as an extension of RS object detection [3]. Recently, RSVG has received growing attention owing to the following reasons. On the one hand, understanding RS images at the scene level becomes increasingly important as the resolution increases [4], [5], [6]. Detecting and locating the specific objects within RS images is an essential component of scene interpretation. On the other hand, the development of cross-modal learning techniques provides favorable conditions for RSVG [7], [8], [9]. Therefore, RSVG plays an essential role in many applications, such as image retrieval [10], [11], [12], [13], scene classification [14], [15], [16], [17], hyperspectral image classification [18], [19], visual question answering [20], [21], and semantic segmentation [22], [23].

The first RSVG method may be the GeoVG proposed in [24]. It contains a text encoder, an image encoder with adaptive region attention, and a fusion module. They focus on learning textual features with the consideration of geospatial relationships, extracting rich visual features, and fully integrating the textual and visual features. By combining them, the positive results can be achieved. Inspired by GeoVG, many RSVG models have been proposed [1], [2], [3], [25]. Most are accustomed to using visual convolutional neural networks (CNNs) and language networks [3] to extract visual and textual features from RS images and textual descriptions. Then, they establish relationships between the visual and textual information by specific schemes, such as multiscale cross-attention [1], [2] and multistage synergistic aggregation [25], to achieve the localization of specific targets. Like the standard cross-modal technologies, RSVG needs to align visual and textual features for the final results. Unlike common cross-modal approaches, RSVG places equal emphasis on object- and scene-level clues, emphasizing the need for object recognition and understanding.

Despite the remarkable results of previous works, challenges still deserve further research. First, due to the specific characteristics of RSVG, the expected visual features should

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contain crucial textual clues hidden in descriptions. Nevertheless, previous methods [2], [25] always explore visual and textual features separately but overlook mining the mutual influence between them. Although some works (e.g., [1]) have attempted to use textual features to assist in learning visual features, the results often fall short of our expectations due to the issue of feature misalignment. Second, generating exact bounding boxes does not receive the attention it deserves. Directly regressing target bounding boxes by a once-for-all decoder and a grounding head may result in an inadequate understanding of the semantic information, leading to mislabeling or missed detections [1], [2], [25]. Some approaches (e.g., [3]) have proposed more effective bounding box regression methods; however, they primarily focus on visual understanding and overlook the uniqueness of RSVG, especially in terms of textual information comprehension. Last but not least, mining interactions between visual and text features during the target localization is insufficient. This would lead to the imprecise alignment of text-referred objects and introduce ambiguity in target localization. Achieving comprehensive alignment between visual and textual features to enable precise localization of text-specified targets remains a fundamental challenge in RSVG tasks.

To address the challenges mentioned above, this article proposes a new RSVG method named the language-guided progressive query refinement with diffusion model for visual grounding (LPQR-DVG). It mainly contains a multiscale text-guided feature learning module (MTFLM) and an iterative query localization module (IQLM). MTFLM aims to obtain visual features that perceive textual messages, which enclose a textual backbone, a text-guided visual feature learning block, and a transformer encoder. Their functions are to learn text features, mine text-aware visual features, and capture global contexts within RS images and text descriptions. IQLM is developed to generate accurate bounding boxes and thoroughly complete the visual–text interaction for detecting and locating explicit targets. To these ends, IQLM introduces a RSVG-oriented reversed U-Net with progressive query refinement (PQR) units. Through constant interaction among queries, visual and text features, comprehensive visual–text alignment, and accurate localization, corresponding to textual descriptions can be achieved. Finally, to ensure the queries comprehensively integrate both visual and textual information, we introduce a progressive diffusion learning strategy that continuously refines target localization through multistep denoising.

The main contributions of this article are summarized as follows.

1) A new RSVG method named LPQR-DVG is proposed. By incorporating text-guided learning during the visual feature extraction stage and introducing query refinement cross-attention, our method progressively aligns visual and textual features. Furthermore, by employing the diffusion paradigm to refine localization iteratively, accurate identification of text-referred targets can be achieved.
2) The MTFLM module is introduced to provide sufficient text guidance throughout the visual feature extraction process for detailed visual–text understanding. Specifically, we project text features into multiscale visual feature learning, enabling cross-attention interaction between text and visual features at different spatial resolutions. This way, visual features can capture semantic information that is more closely related to textual descriptions, allowing for more accurate object localization.
3) The IQLM module, composed of RSVG-oriented reversed U-Net with PQR units, is introduced to accomplish cross-modal feature alignment and iterative target localization jointly. PQR units can not only integrate textual and visual information into the queries through cross-modal interaction, but also thoroughly align visual and text features at each unit. Thus, the queries can sufficiently capture the target information in the visual features referred to by the text.
4) The progressive diffusion learning strategy is employed to train our model and ensure the accuracy of the results. Specifically, the strategy progressively denoises the random bounding box coordinates through iterative interactions with visual and textual features in the RSVG-oriented reversed U-Net. Through iterative denoising steps, the random coordinates can gradually transfer to semantically relevant regions, ultimately generating precise target coordinates as the final prediction.

II. Related Work

A. Visual Grounding in Natural Scenes

The existing natural VG models can generally be classified into two- and one-stage methods. Two-stage methods [26], [27] typically employ an object detector first to identify multiple candidate boxes in a natural image. Then, the optimal box will be selected by the ranking schemes according to the textural information, thereby localizing the target objects. For example, Yang et al. [26] proposed a dynamic graph attention network for VG. First, it obtains multiple object features in an image by using ground truth objects and an image backbone or employing a pre-trained object detector to detect objects. Meanwhile, text features are obtained from word embedding. Then, a dynamic graph attention network is used to build a graph with objects and their relationships, incorporating the object information and relations described in the text into the graph. Finally, the target is selected by computing the matching scores between each object and the textual expressions.

In contrast, one-stage approaches directly estimate the positions of the referring objects without generating a set of candidate regions. Initially, CNNs were popular due to their strong feature learning capacity. These CNN-based models integrate an object detector with a carefully designed VG module [28], [29], [30], [31]. They use text encoders and CNNs to facilitate cross-modal interaction between visual and text features. For instance, to better understand spatial positional relationships in textual information, Huang et al. [29] proposed LBYL-Net. It uses pre-trained visual and text encoders to extract visual and text features, followed by a fusion module with an affine transformation on visual

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features guided by text to perform cross-model interaction. Then, a landmark feature convolution module is introduced to integrate contextual information and gather visual features from different directions. Finally, a localization module with a detection head is used to localize the target. Subsequently, owing to the ability in long-distance relationship capture and cross-model learning, the transformer became popular in VG tasks [32], [33], [34]. Yang et al. [35] proposed a VG model named VLTVG. It first leverages the normal transformer-based visual and textual encoders to map images and texts into the visual and textual embeddings. Then, based on the multihead self-attention mechanism, a visual-linguistic verification module, a language-guided context encoder, and a multistage cross-modal decoder are developed to generate text-conditioned visual features and achieve the target localization. Another set of transformer-based methods was introduced in [36] and [37], which view VG as a direct set prediction problem [38]. Like VLTVG, they adopt transformers to capture valuable visual and textual representations and mine their relations. Unlike VLTVG, they complete the target location using random and multiple query strategies. Recently, diffusion models [39] have attracted researchers' attention since their paradigm enables iterative reasoning for visual–text feature alignment as well as progressive target localization [40], [41]. Chen and Li [41] applied generative diffusion models to VG and proposed a language-guided diffusion model, LG-DVG. It incorporates iterative visual–text reasoning and alignment while progressively refining bounding boxes for query-region matching. LD-DVG performs object localization by denoising a set of noisy boxes generated by the visual encoder. These boxes interact with text features extracted by the text encoder, progressively denoise and localize the target object.

B. RS Visual Grounding

Current research on RSVG predominantly focuses on one-stage approaches. Sun et al. [24] first introduced VG into the RS community. Considering the differences between RS and natural images, they first collected a dataset for RSVG tasks. Then, a convolutional-based model, GeoVG, was developed to achieve target localization. Given complex geographical spatial relationships and the absence of visually prominent objects in RS images, GeoVG achieves better localization of RS images through a language encoder with a geospatial relation graph, a visual encoder with large-scale region attention, and a fusion module to localize targets. Another dataset (i.e., DIOR-RSVG) was constructed in [2]. Compared with the above dataset, the RS images within DIOR-RSVG have higher spatial resolution. Also, the volume of DIOR-RSVG is larger. Also, a transformer-based method named MGVLF was proposed in this work to address the challenges of large-scale variations, cluttered backgrounds, and significant scale changes in RS images. First, multiscale visual features and text features are extracted from pre-trained visual and text encoders. Following this, MGVLF employs cross-modal and self-attention transformers to enable interaction between multiscale visual and textual features while filtering out irrelevant noise to learn more discriminative representations. A fusion transformer and a localization module are further introduced to obtain the predicted target location. Although MGVLF performs well, the insufficient feature interactions and inaccurate coordinate predictions still weaken its behavior. As a result, Li et al. [1] proposed a new transformer-based model called LPVA. Apart from multiscale visual–text alignment after feature extraction, it also pays attention to the interaction between visual and text features during the feature extraction process, where text features are introduced to progressively refine visual features in the visual backbone through an attention module. After that, a multilevel feature enhancement decoder is adopted to perform visual–text interaction and fusion and achieve target localization through a localization module.

The above transformer-based methods focus on the design of visual–text alignment modules. However, they are not good at learning the target-oriented positional representation. To mitigate this issue, Lan et al. [3] proposed LQVG, inspired by DETR [38], which has a query-based transformer framework. It first extracts multiscale visual and text features using pre-trained image and text encoders, respectively. Subsequently, a multiscale cross-modal alignment module is developed to align visual and text features at different scales. Next, sentence-level text features are used as language queries to retrieve and aggregate the referred object representation from multiscale visual features in the transformer decoder. Finally, a localization module transforms these queries into target coordinates and category information. Meanwhile, considering the single fusion strategy in RSVG may limit the model's ability to learn the complex associations between textual semantics and multiscale visual information, a new model named MSAM was developed in [25], in which a generative paradigm is harnessed to directly generate a discrete sequence of coordinates in an auto-regressive manner.

C. Distinctive Features of LPQR-DVG

The key ideas of our work are the introduction of language information and the diffusion module. Many methods have been proposed based on these two techniques, such as [1], [2], [3], [25], [41], and [42]. However, the proposed LPQR-DVG differs from them and has many of its own properties. For the inclusion of linguistic clues, the main difference between LPQR-DVG and the existing ones is the way of utilizing the textual knowledge. Differing from [1], which relies on generating language-adaptive weights and biases to guide visual feature extraction, and unlike [2] and [3], which exclude entirely textual messages during the visual feature extraction stage, our LPQR-DVG introduces MTFLM to achieve deeper visual–textual interaction during the visual feature extraction stage. Specifically, MTFLM projects textual features into each layer of the visual feature extractor, followed by direct interaction through cross-attention mechanisms. Subsequently, IQLM and the diffusion paradigm are incorporated to enable visual features to progressively adapt to target locations specified by the text, thereby achieving precise localization. In this manner, not only can the visual representations be learned with the guidance of textual clues, but also the final localization results are sensitive to textual descriptions.

Existing diffusion-based methods [41], [42] primarily focus on natural scene understanding, often requiring multiple anchor points or pre-trained diffusion models to locate single or multiple simple objects. When applying them to RS images,

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numerous challenges arise due to the complex object relationships and size variations of specific targets. For example, objects in RS images often lack clear boundaries and exhibit intricate spatial relationships. Also, these targets are diverse in scale, ranging from small isolated objects (e.g., "tiny vehicle") to large complex areas (e.g., "expressway service area"). Our LPQR-DVG is developed to overcome these challenges. In detail, we introduce a progressive diffusion-denoising process within the IQLM, where noisy queries interact iteratively with both visual and textual information. This enables the queries to capture complex object characteristics and spatial contextual relationships gradually. Moreover, unlike methods that rely on multiple candidate proposals, we formulate the diffusion process with a single target query. This design allows LPQR-DVG to focus on precisely localizing referred objects, thereby reducing the recognition complexity associated with extreme variation in object scales.

In addition, the adoption of diffusion modeling stems from its distinct advantages over alternative paradigms for bounding box generation. Unlike nongenerative detectors that produce deterministic coordinates through regression [1], [3], [25], diffusion models probabilistically refine predictions through iterative denoising. Through iterative denoising steps, the random coordinates can gradually transfer to semantically relevant regions, ultimately generating precise target coordinates as the final prediction. Meanwhile, compared to other generative frameworks, diffusion models avoid the instability of GANs' [43] adversarial training and the tendency of VAEs' [44] to produce blurry outputs. Critically, the reverse diffusion process inherently emulates spatial cognition by progressively refining positional certainty through iterative denoising. This architecture proves particularly advantageous for language-guided localization where targets exhibit substantial inter-instance appearance variance.

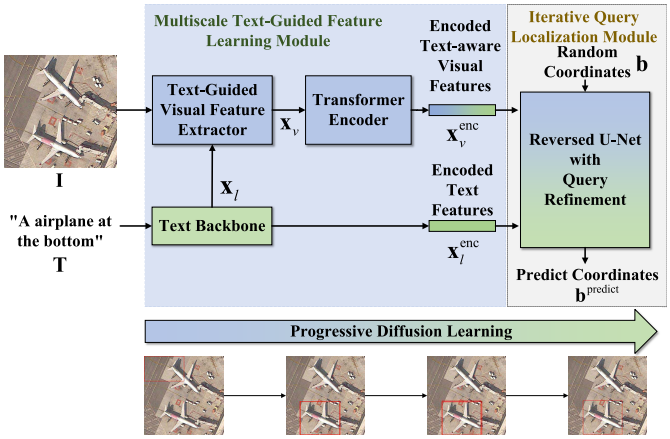


Fig. 1. Pipeline of LPQR-DVG. It consists of an MTFLM and an IQLM. MTFLM aims to extract and integrate visual and text features effectively. IQLM pays attention to deeply interacting with two types of features and locating the targets precisely. In addition, a progressive diffusion learning strategy is introduced to train LPQR-DVG for accurate visual grounding.

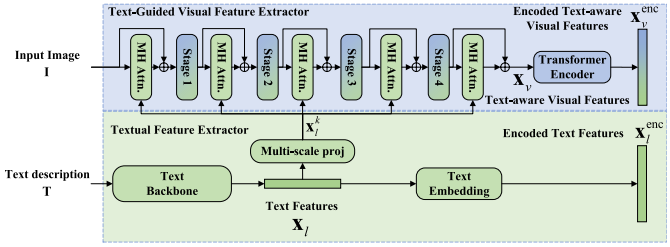


Fig. 2. Illustration of MTFLM. It consists of a text-guided visual feature extractor and a textual feature extractor. The textual feature extractor is responsible for extracting text features and projecting them to multiple scales to match the multiscale visual features produced by the text-guided visual feature extractor, as well as the final visual output. The text-guided visual feature extractor leverages textual guidance to extract text-aware visual features and employs a transformer encoder to capture global contextual information.

III. Methods

A. Overview

The overall framework of LPQR-DVG is illustrated in Fig. 1. Given an RS image $\mathbf{I} \in \mathbb{R}^{C\times H\times W}$ and its corresponding textual description $\mathbf{T}$ with arbitrary length, LPQR-DVG first uses MTFLM to extract text-aware visual features $\mathbf{x}_v^{\text{enc}}$ and textual features $\mathbf{x}_l^{\text{enc}}$. Then, an IQLM is introduced to enhance the interaction between $\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$, while achieving precise target localization. Finally, we introduce a progressive diffusion learning strategy to fully train LPQR-DVG with the consideration of the visual–text feature alignment and accuracy of bounding boxes. Now, we will explain MTFLM, IQLM, and the progressive diffusion learning strategy in detail. Here, C, H, and W denote the channels, height, and width of the RS image, respectively.

B. Multiscale Text-Guided Feature Learning Module

As shown in Fig. 2, MTFLM encloses a text-guided visual feature extractor and a textual feature extractor. The textural feature extractor comprises a BERT [45] model, multiscale projection layers, and a text embedding layer. BERT is used to extract semantic embeddings $\mathbf{x}_l \in \mathbb{R}^{s\times d_l}$ from the input texts, where s, d_l represents total sequence length and dimensions. Multiscale projection layers are dedicated to mapping $\mathbf{x}_l$ into the immediate textual features with different dimensions. These immediate features are responsible for transmitting semantic knowledge to the visual features. For clarity, we record them as $\{\mathbf{x}_l^k, k = 0, 1, 2, 3, 4\}$, where k represents the kth stage of the visual encoder. The text embedding layer aims to map $\mathbf{x}_l$ into the encoded text features $\mathbf{x}_l^{\text{enc}} \in \mathbb{R}^{s\times d}$, where d donates the dimensions the encoded text features. Here, the multiscale projection layers and the text embedding layer are composed of fully connected layers.

The backbone of the text-guided visual feature extractor is ResNet [46]. For the specific RSVG tasks, the expected visual features should contain multiscale clues hidden in the RS image $\mathbf{I}$ and be aware of the semantics of the input texts. To this end, we embed the immediate textual features $\mathbf{x}_l^k$ achieved by the textual feature extractor into different stages of ResNet by using a simple multihead cross-attention mechanism [47] and the residual connection. This step corresponds to the kth stage of modified ResNet and it can be formulated as

$$\hat{\mathbf{x}}_v^k = \mathbf{x}_v^k + \text{Reshape}\left(\text{Attn}\left(\text{Reshape}\left(\mathbf{x}_v^k\right), \mathbf{x}_l^k, \mathbf{x}_l^k\right)\right) \quad (1)$$

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where $\mathbf{x}_v^k \in \mathbb{R}^{c_k \times h_k \times w_k}$ and $\hat{\mathbf{x}}_v^k \in \mathbb{R}^{c_k \times h_k \times w_k}$ denote the outputs of the kth stage of ResNet and modified ResNet, Attn $(\cdot)$ indicates the function of the multihead cross-attention mechanism, and Reshape $(\cdot)$ means the conversion function between 2-D and 1-D data. Here, the channels, heights, and widths of $\mathbf{x}_v^k$ and $\hat{\mathbf{x}}_v^k$ are c_k, h_k, and w_k. As a result, the text-aware visual features $\mathbf{x}_v \in \mathbb{R}^{c \times h \times w}$, i.e., the final outputs of the modified ResNet, can be produced, which contain local visual clues and textual intelligence. Global context relations should also be explored to further understand the complex contents of RS images. Therefore, the transformer encoder [38] is embedded at the top of the modified ResNet. After passing $\mathbf{x}_v$ through this encoder, we can get the encoded text-aware visual features $\mathbf{x}_v^{\text{enc}} \in \mathbb{R}^{s_v \times d}$, where $s_v = h \times w$ is the sequence length of $\mathbf{x}_v^{\text{enc}}$, and d means the dimension of $\mathbf{x}_v^{\text{enc}}$. In this article, we set $s_v = 625$ and $d = 256$.

C. Iterative Query Localization Module

To jointly perform feature decoding and coordinate localization, we propose IQLM based on the reversed U-Net [48], [49], [50]. IQLM takes random coordinates $\mathbf{b} \in \mathbb{R}^{1\times 2\times 2}$ as inputs, aiming to predict the bounding box coordinates $\mathbf{b}^{\text{predict}} \in \mathbb{R}^{1\times 2\times 2}$ in the RS image corresponding to the text description. Following [40], a scaling function and a 1×1 convolutional layer are used to normalize $\mathbf{b}$ to query $\mathbf{q} \in \mathbb{R}^{c\times 2\times 2}$ at the beginning of IQLM.

The standard reversed U-Net consists of four encoding blocks, one middle block, and four decoding blocks. The encoding blocks aim to extract multiscale features from the input query and capture detailed information essential for target localization. The middle block processes high-dimensional feature representations and refines them into more precise feature maps. The decoding blocks are used to generate localization outputs based on these refined features. In addition, the up- and down-sampling operations are embedded in the encoding and decoding blocks to control the data sizes. The encoding, middle, and decoding blocks are all composed of the residual unit, whose structure is the same as the U-Net residual part in Fig. 3. It primarily consists of two convolutional layers, two group normalization layers, and a normalization and projection layer, with all components connected in a residual manner. Suppose the input feature of the ith residual unit is $\mathbf{q}_i$, the output $\mathbf{q}_i^{\text{emb}}$ can be obtained by

$$\mathbf{q}_i^{\text{emb}} = \mathbf{q}_i + \text{Conv}\left(\text{GN}\left(\text{Conv}\left(\text{GN}\left(\mathbf{q}_i\right)\right)\right)\right) \tag{2}$$

where Conv $(\cdot)$ and GN $(\cdot)$ denote the functions of the convolutional and group normalization layers. In the standard reversed U-Net, each encoding block contains two residual units, the middle block incorporates one residual unit, and each decoding block encompasses three residual units.

Although the standard reversed U-Net is able to map the random query $\mathbf{q}$ into the bounding box coordinates $\mathbf{b}^{\text{predict}}$, it does not take the visual and textual features ($\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$) learned by MTFLM into account. This makes $\mathbf{b}^{\text{predict}}$ less related to visual and textual information, so it cannot meet the requirement for RSVG. To address this issue, it is essential to incorporate both visual and textual information during the coordinate learning process, as well as to facilitate a thorough interaction between visual and textual features. Therefore, we construct an RSVG-oriented reversed U-Net, in which a PQR unit is developed to replace the residual unit in the standard reversed U-Net. Let us take the ith PQR unit as an example to explain its structure and function in detail. As shown in Fig. 3, the PQR unit consists of a U-Net residual part and a query refinement part. The U-Net residual part aims to capture hierarchical feature representations from the input data, and the query refinement part focuses on refining them by employing the interactions between visual and textural features, ultimately achieving accurate target localization. For clarity, the input and output of the ith PQR unit are recorded as $\mathbf{q}_i$ and $\mathbf{q}_i^{\text{refine}}$. When $\mathbf{q}_i$ is input, the PQR unit first maps it into $\mathbf{q}_i^{\text{emb}}$ using the U-Net residual part according to (2). Then, the query refinement part is adopted to enable $\mathbf{q}_i^{\text{emb}}$ to interact with $\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$, thereby acquiring the target referred by textual information. Meanwhile, $\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$ can be progressively aligned by the multihead attention mechanism [47], which enables $\mathbf{x}_v^{\text{enc}}$ to highlight messages related to targets referred by $\mathbf{x}_l^{\text{enc}}$, thereby enhancing its discriminative ability.

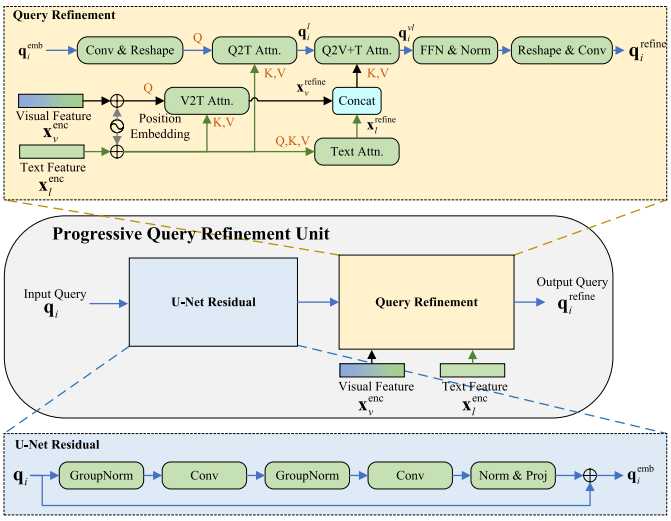


Fig. 3. Structure of each PQR unit in RSVG-oriented reversed U-Net. It consists of a U-Net residual part and a query refinement part. The U-Net residual part is designed to capture hierarchical feature representations from the input query, while the query refinement part facilitates visual–text feature alignment and enables interactions between the query and both visual and text features.

As presented in Fig. 3, the query refinement part first adds learnable position embeddings to $\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$, allowing the attention mechanism to incorporate spatial and semantic position information. After that, a 1×1 convolution layer and reshape operation are used to project $\mathbf{q}_i^{\text{emb}}$ into 2-D data, ensuring its dimensions align with $\mathbf{x}_v^{\text{enc}}$ and $\mathbf{x}_l^{\text{enc}}$.

Next, we introduce a multihead self-attention layer and three multihead cross-attention layers to improve query and various features by interacting with their latent information. Finally, we obtain the final output $\mathbf{q}_i^{\text{refine}}$ of the ith PQR unit with a reshape operation and a 1×1 convolution layer. The whole process of the query refinement part can be expressed as

$$\mathbf{x}_v^{\text{refine}} = \text{CAttn}\left(\mathbf{x}_v^{\text{enc}}, \mathbf{x}_l^{\text{enc}}, \mathbf{x}_l^{\text{enc}}\right)$$
$$\mathbf{x}_l^{\text{refine}} = \text{SAttn}\left(\mathbf{x}_l^{\text{enc}}, \mathbf{x}_l^{\text{enc}}, \mathbf{x}_l^{\text{enc}}\right)$$

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$$\begin{aligned}\mathbf{q}_i^l &= \mathrm{CAttn}\left(\mathrm{Reshape}\left(\mathrm{Conv}\left(\mathbf{q}_i\right)\right), \mathbf{x}_l^{\mathrm{refine}}, \mathbf{x}_l^{\mathrm{refine}}\right)\\ \mathbf{q}_i^{\mathrm{vl}} &= \mathrm{CAttn}\left(\mathbf{q}_i^l, \left[\mathbf{x}_l^{\mathrm{refine}}, \mathbf{x}_v^{\mathrm{refine}}\right], \left[\mathbf{x}_l^{\mathrm{refine}}, \mathbf{x}_v^{\mathrm{refine}}\right]\right)\\ \mathbf{q}_i^{\mathrm{refine}} &= \mathrm{Conv}\left(\mathrm{Reshape}\left(\mathbf{q}_i^{\mathrm{vl}}\right)\right)\end{aligned} \tag{3}$$

where Conv(·) represents convolutional layers, SAttn(·) and CAttn(·) mean the multihead self- and cross-attention layers, Reshape(·) indicates the conversion function between 3-D and 2-D data, and [·] denotes feature concatenation along the sequence dimension. There is another point we want to touch on, that is, $\mathbf{q}_i$ is varied according to its position in the RSVG-oriented reversed U-Net. In the first encoding block, $\mathbf{q}_i$ is equal to the initial query $\mathbf{q}$. In the following encoding and mid blocks, $\mathbf{q}_i$ is the output of the $(i-1)$th PQR unit $\mathbf{q}_{\mathbf{i-1}}^{\mathrm{refine}}$. In decoding blocks, $\mathbf{q}_i$ can be obtained from the concatenation of $\mathbf{q}_{\mathbf{i-1}}^{\mathrm{refine}}$ and the output of corresponding jth PQR encoding unit $\mathbf{q}_{\mathbf{j}}^{\mathrm{refine}}$.

Based on the PQR unit, our RSVG-oriented reversed U-Net is constructed. It consists of four encoding blocks, one middle block, and four decoding blocks. Each encoding block contains two PQR units, the middle block contains one PQR unit, and each decoding block contains three PQR units. The RSVG-oriented reversed U-Net can not only improve the query with the assistance of visual and text features but also thoroughly align different features. Hence, the localization performance can be ensured. Note that the refined query $\mathbf{q}_i^{\mathrm{refine}}$ from the last PQR unit would be projected by a convolutional layer and then passed through a re-signal function to achieve the predicted bounding box $\mathbf{b}^{\mathrm{predict}}$.

D. Diffusion Process and Training Loss

This article introduces a diffusion paradigm to train the proposed LPQR-DVG. It progressively removes noises added to the query $\mathbf{q}$ under the guidance of $\mathbf{x}_v^{\mathrm{enc}}$ and $\mathbf{x}_l^{\mathrm{enc}}$ to push the model achieves better visual–text alignment and improved referred target localization. To complete the diffusion process, the sinusoidal positional embedding and MLP layers are used to generate time steps $t_{\mathrm{emb}}^i \in \mathbb{R}^{c_i}$, and these time steps would be added to the input query after the first convolutional layer of the U-Net residual part in each PQR unit. Here, c_i represents the channels in the ith PQR unit.

The schematic of the diffusion procedure is displayed in Fig. 4. In the training stage, the ground truth bounding box $(\mathbf{x}, \mathbf{y}, \mathbf{w}, \mathbf{h})$ is first used as the initial coordinate input $\mathbf{b}_0 \in \mathbb{R}^{1\times2\times2}$, where $(\mathbf{x}, \mathbf{y})$ denotes the center coordinates of the bounding box, and $(\mathbf{w}, \mathbf{h})$ represents its width and height, respectively. Consistent with IQLM, a signal function is used to scale the input bounding box, resulting in the signal box $\mathbf{b}_{\mathbf{0}}^{\mathrm{signal}}$. Then, following the diffusion paradigm in DDPM [51], a Markovian chain of diffusion forward process is defined by gradually adding noise to the initialized coordinates, which can be defined as:

$$\begin{aligned}\bar{\alpha}_t &= \prod_{s=0}^{t} \alpha_s = \prod_{s=0}^{t} \beta_s\\ \mathbf{b}\left(\mathbf{b}_t^{\mathrm{signal}} \middle| \mathbf{b}_0^{\mathrm{signal}}\right) &= N\left(\mathbf{b}_t^{\mathrm{signal}} \middle| \sqrt{\bar{\alpha}_t}\mathbf{b}_0^{\mathrm{signal}}, (1-\bar{\alpha}_t)I\right)\end{aligned} \tag{4}$$

where β_s represents the noise variance schedule at step s, $\bar{\alpha}_t$ denotes the cumulative product of noise schedule coefficients

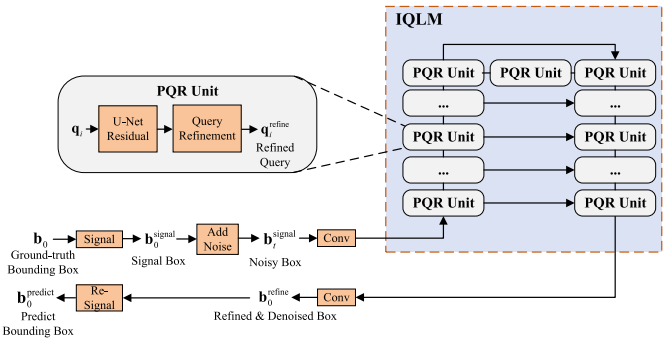


Fig. 4. Schematic of the diffusion procedure for producing the location bounding box. Specifically, the signal and re-signal functions are employed to scale the bounding box to an appropriate range. By adding noise to the signal box and enabling continuous interactions with visual and textual features within IQLM, a denoised bounding box can be obtained. Finally, the predicted bounding box is produced as the final output.

at step t, $\mathbf{b}_0^{\mathrm{signal}}$ implies the initial signal box, and $\mathbf{b}_t^{\mathrm{signal}}$ illustrates the latent noisy box at diffusion step t, which can be obtained via sampling from the conditional normal distribution defined above. This forward process transforms the signal box $\mathbf{b}_{\mathbf{0}}^{\mathrm{signal}}$ to a latent noisy box $\mathbf{b}_t^{\mathrm{signal}}$ by adding noise at a time t to $\mathbf{b}_0^{\mathrm{signal}}$, where $t \in \{0, 1, \ldots, T\}$. In this article, we set $T = 1000$. Next, $\mathbf{b}_t^{\mathrm{signal}}$ is passed through a convolutional projection layer and fed into the IQLM. As mentioned in Section III-C, after we obtain the PQR last unit's output $\mathbf{q}_i^{\mathrm{refine}}$ from IQLM, a convolutional projection layer transforms $\mathbf{q}_i^{\mathrm{refine}}$ into the refined and denoised bounding box $\mathbf{b}_0^{\mathrm{refine}}$. Finally, $\mathbf{b}_0^{\mathrm{refine}}$is passed through a re-signal function to obtain the predicted target bounding box $\mathbf{b}_0^{\mathrm{predict}}$. In the inference stage, following the diffusion paradigm, we first randomly initialize a noisy box $\mathbf{b}_0 \in \mathbb{R}^{1\times2\times2} \sim \mathcal{N}(0, 1)$, and then normalize it to $\mathbf{b}_0^{\mathrm{signal}}$. Then, the signal box $\mathbf{b}_0^{\mathrm{signal}}$ is input into the RSVG-oriented reversed U-Net. By progressively refining the denoised query, the final output $\mathbf{b}_0^{\mathrm{predict}}$ can be obtained.

The loss function $\mathcal{L}_{\mathrm{train}}$ in this article consists of two parts, i.e., the box localization loss $\mathcal{L}_{\mathrm{train}}^{\mathrm{box}}$ and auxiliary box refinement loss $\mathcal{L}_{\mathrm{train}}^{\mathrm{aux}}$. $\mathcal{L}_{\mathrm{train}}^{\mathrm{box}}$ is designed to jointly optimize precise coordinate regression and geometric consistency in the diffusion process, which can be defined as

$$\mathcal{L}_{\mathrm{train}}^{\mathrm{box}} = \lambda_1 \mathcal{L}_{\mathrm{smooth\text{-}L_1}}\left(\mathbf{b}_0, \mathbf{b}_0^{\mathrm{predict}}\right) + \lambda_2 \mathcal{L}_{\mathrm{GIOU}}\left(\mathbf{b}_0, \mathbf{b}_0^{\mathrm{predict}}\right) \tag{5}$$

where $\mathcal{L}_{\mathrm{smooth\text{-}L_1}}(\cdot)$ and $\mathcal{L}_{\mathrm{GIOU}}(\cdot)$ denote the smooth-L1 loss and generalized intersection over union (GIoU) loss [52], $\mathbf{b}_0$ and $\mathbf{b}_0^{\mathrm{predict}}$ indicate the ground truth coordinate and output denoised predict bounding box, and λ_1 and λ_2 mean the balance coefficient. $\mathcal{L}_{\mathrm{train}}^{\mathrm{aux}}$ is developed to enhance feature learning through deep supervision, whose definition is

$$\begin{aligned}\mathcal{L}_{\mathrm{train}}^{\mathrm{aux}} &= \lambda_1 \mathcal{L}_{\mathrm{smooth-}L_1}\left(\mathbf{b}_0, \mathbf{b}_0^{\mathrm{aux}}\right) + \lambda_2 \mathcal{L}_{\mathrm{GIOU}}\left(\mathbf{b}_0, \mathbf{b}_0^{\mathrm{aux}}\right)\\ \mathbf{b}_0^{\mathrm{aux}} &= \mathrm{MLP}\left(\mathrm{avgpool}\left(\mathbf{q}_i^{\mathrm{refine}}\right)\right)\end{aligned} \tag{6}$$

where $\mathbf{b}_0^{\mathrm{aux}}$ is the auxiliary predict bounding box of each PQR unit, avgpool(·) and MLP(·) illustrate the function of average pooling and multilayer perceptron, and λ_1 and λ_2 also mean the balance coefficient. In this article, we fix $\lambda_1 = 5$ and $\lambda_2 = 2$ according to [3]. More details related to $\mathcal{L}_{\mathrm{train}}^{\mathrm{aux}}$ can be

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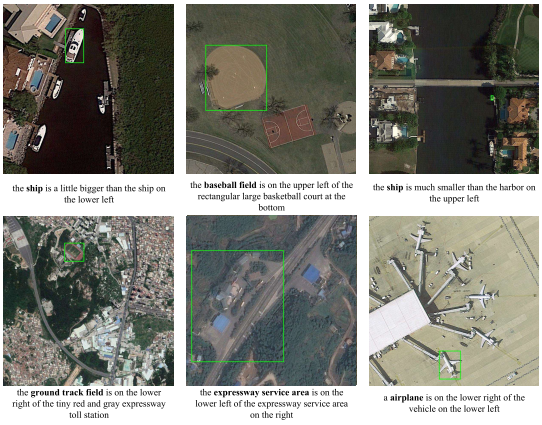


Fig. 5. Visual examples and their textual descriptions. Six RS scenes are randomly selected from DIOR-RSVG, and they are located at the top of each cube. Their corresponding textual descriptions are exhibited at the bottom of each block. The green boxes in these RS scenes are the ground truths.

found in DETR [38]. Finally, the loss function for our model is formulated as

$$\mathcal{L}_{\text{train}} = \mathcal{L}^{\text{box}}_{\text{train}} + \alpha \mathcal{L}^{\text{aux}}_{\text{train}} \tag{7}$$

where α is a hyperparameter that balances the contributions of the two terms. In this article, we set $\alpha = 1$ empirically.

IV. Experiments

A. Datasets Description

To testify to the effectiveness and robustness of our method, we select two popular datasets, including DIOR-RSVG [2] and OPT-RSVG [1]. The DIOR-RSVG dataset was constructed by researchers from Northwestern Polytechnical University in 2023. It is an extension of the DIOR dataset [53]. DIOR-RSVG contains 17 402 high-resolution RS scenes. Their spatial sizes are 800×800, and their spatial resolutions range from 0.5 to 30 m. There are 38 320 textual descriptions in DIOR-RSVG corresponding to referred objects across 20 classes, such as airplane and basketball court. The average length of these descriptions is 7.47 words. Some visual examples and their textual descriptions are shown in Fig. 5. Following [2], we fix the ratio of training, validating, and testing sets as 7:1:2. In other words, 26 991 (70%), 3829 (10%), and 7500 (20%) image-language pairs are randomly selected to construct the training, validation, and testing sets.

The OPT-RSVG dataset was constructed by researchers from Xidian University in 2024. It contains 25 452 RS scenes, and their spatial sizes vary from 152×162 to $8935 \times 10\,569$. There are 48 952 text descriptions across 14 object categories, including crossroads, bridges, and others. The average length of these text descriptions is 10.10 words. Some visual examples and their textual descriptions are shown in Fig. 6. Following [1], 19 580 (40%), 4895 (10%), and 24 477 (50%) image-language pairs are randomly selected to construct the training, validation, and testing sets. Compared to DIOR-RSVG, OPT-RSVG presents greater challenges due to its extreme variations in target scales, increased similar objects, diverse viewing angles, and complex background noise.



Fig. 6. Visual examples and their textual descriptions. Six RS scenes are randomly selected from OPT-RSVG, and they are located at the top of each cube. Their corresponding textual descriptions are exhibited at the bottom of each block. The green boxes in these RS scenes are the ground truths.

B. Experimental Settings

All experiments are conducted on a workstation equipped with a GeForce RTX 3090 with 24 GB of memory. The parameters of the ResNet backbone, transformer encoder, and textual backbone in MTFLM are initialized using the pre-trained weights in DETR [38] and BERT [45]. Other parts of our LPQR-DVG are initialized randomly. The channel multipliers in the RSVG-oriented reversed U-Net are used to scale the number of channels at each layer. Specifically, we initialize the model with 32 base channels and employ multipliers $[1, 2, 2, 2]$ to determine how the channel count evolves through each block of the RSVG-oriented reversed U-Net. The AdamW optimizer [54] is employed to update the LPQR-DVG's parameters. Its initial learning rates are set as 1×10^{-5} and 1×10^{-4} for MTFLM and IQLM. The learning rate is reduced by a factor of 0.1 at the 60th epoch. The total number of training epochs is 100 (equivalent to 300 000 diffusion steps), and the batch size is fixed at 9. The inference query sampling steps in DDPM [51] are set to ten by default.

Three assessment criteria are adopted to evaluate LPQR-DVG's performance [1], [2], [3], including Precision@X (Pr @X), mean intersection over union (mIoU), and cumulative intersection over union (cIoU). Before defining them, we first introduce intersection over union (IoU). It calculates the overlaps between the predicted boxes and the ground truth boxes, which can be formulated as $\text{IoU}_i = \mathbf{I}_i/\mathbf{U}_i$, where IoU_i represents the IoU of ith sample, $\mathbf{I}_i$ is the area of overlap between the predicted bounding box and the ground truth bounding box, $\mathbf{U}_i$ is the total area covered by both the predicted and ground truth bounding boxes. Based on IoU, Pr @X can be defined as the percentage of IoU overlap exceeding a threshold X, mIoU can be interpreted as the average IoU values across all testing samples, and cIoU can be described as the cumulative IoU values over all predictions. Note that the threshold X ranges from 0.5 to 0.9 in the following experiments. Among them, the definitions of mIoU and cIoU are

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{I}_i/\mathbf{U}_i\,, \quad \text{cIoU} = \sum_{i=1}^{N} \mathbf{I}_i / \sum_{i=1}^{N} \mathbf{U}_i \tag{8}$$

where N represents the total number of input samples.

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TABLE I
PERFORMANCE OF DIFFERENT METHODS COUNTED ON DIOR-RSVG

Methods	Venue	Visual Encoder	Language Encoder	$Pr@0.5$ (%)	$Pr@0.6$ (%)	$Pr@0.7$ (%)	$Pr@0.8$ (%)	$Pr@0.9$ (%)	$mIoU$	$cIoU$
CNN-based:										
FAOA [28]	*ICCV'19*	DarkNet-53	BERT	60.69	52.28	39.53	21.29	4.32	0.4925	0.5768
FAOA [28]	*ICCV'19*	DarkNet-53	LSTM	63.41	55.18	41.60	23.72	5.24	0.5165	0.6042
ReSC [30]	*ECCV'20*	DarkNet-53	BERT	70.38	64.76	54.02	36.24	10.06	0.5834	0.6883
LBYL-Net [29]	*CVPR'21*	DarkNet-53	BERT	70.84	64.51	53.83	35.61	8.52	0.3276	0.7944
LBYL-Net [29]	*CVPR'21*	DarkNet-53	LSTM	72.53	66.79	56.97	39.89	11.17	0.4819	0.8042
transformer-based:										
TransVG [33]	*ICCV'21*	ResNet-50	BERT	70.73	65.04	55.73	41.72	17.28	0.6117	0.7548
VLTVG [35]	*CVPR'22*	ResNet-50	BERT	74.68	70.49	63.55	50.91	27.69	0.6473	0.7484
MGVLF [2]	*TGRS'23*	ResNet-50	BERT	72.92	68.13	60.08	47.55	24.25	0.6321	0.7490
LPVA [1]	*TGRS'24*	ResNet-50	BERT	75.21	70.14	62.12	48.80	24.73	0.6504	0.7898
LPQR-DVG (Ours)	-	ResNet-50	BERT	**78.69**	**74.27**	**66.83**	**54.45**	**32.37**	**0.6844**	**0.8055**

TABLE II
PERFORMANCE OF DIFFERENT METHODS COUNTED ON OPT-RSVG

Methods	Venue	Visual Encoder	Language Encoder	$Pr@0.5$ (%)	$Pr@0.6$ (%)	$Pr@0.7$ (%)	$Pr@0.8$ (%)	$Pr@0.9$ (%)	$mIoU$	$cIoU$
CNN-based:										
FAOA [28]	*ICCV'19*	DarkNet-53	BERT	62.68	54.81	41.23	21.60	3.47	0.5065	0.6114
FAOA [28]	*ICCV'19*	DarkNet-53	LSTM	64.14	55.83	41.54	21.76	3.38	0.5180	0.6066
ReSC [30]	*ECCV'20*	DarkNet-53	BERT	73.24	68.06	57.72	37.89	9.48	0.6012	0.6862
LBYL-Net [29]	*CVPR'21*	DarkNet-53	BERT	39.89	36.49	30.42	19.43	5.43	0.1881	0.4374
LBYL-Net [29]	*CVPR'21*	DarkNet-53	LSTM	72.82	68.29	58.67	39.16	8.63	0.5127	0.7407
transformer-based:										
TransVG [33]	*ICCV'21*	ResNet-50	BERT	73.31	67.54	58.17	41.59	14.21	0.6223	0.7157
VLTVG [35]	*CVPR'22*	ResNet-50	BERT	74.67	70.67	63.30	47.87	19.43	0.6376	0.7374
MGVLF [2]	*TGRS'23*	ResNet-50	BERT	73.81	69.09	61.18	46.17	17.78	0.6323	0.7406
LPVA [1]	*TGRS'24*	ResNet-50	BERT	76.43	71.17	61.71	44.59	15.52	0.6456	0.7547
LPQR-DVG (Ours)	-	ResNet-50	BERT	**78.26**	**74.13**	**66.31**	**50.86**	**21.77**	**0.6693**	**0.7705**

C. Comparison With Existing Methods

To evaluate the effectiveness of our proposed method, we compare it against seven existing methods, including FAOA [28], ReSC [30], LBYL-Net [29], TransVG [33], VLTVG [35], MGVLF [2], and LPVA [1]. The first three methods are developed based on CNN, and the last four are transformer-based approaches. Their details, such as visual and textual backbones, can be found in Tables I and II. Here, FAOA and LBYL-Net have two versions due to the different textual encoders. The results of them and our LPQR-DVG counted on DIOR-RSVG and OPT-RSVG are shown in Tables I and II, respectively.

1) Analysis of Results on DIOR-RSVG: From the observation of the results in Table I, it is evident that the performance of all methods is acceptable. The CNN-based approaches estimate the positions of referred objects based on cross-modal representations, and they eliminate candidate proposals typically used in object detection to enhance the efficiency of predicting accurate target regions through an end-to-end paradigm. Among them, the weakest is FAOA. When its language encoder is BERT, its Pr @0.5, Pr @0.9, mIoU, and cIoU values are only 60.69%, 4.32%, 0.4925, and 0.5768, respectively. This is because FAOA's spatial feature encoding cannot establish the complex correspondence between vision and language. To mitigate this issue, ReSC employs a recursive subquery to refine visual features. As a result, its various metrics are significantly higher than those of FAOA. Compared to FAOA and ResC, the behavior of LBYL-NET is more attractive since it employs a landmark-aware feature convolution mechanism, which is beneficial for the model to understand spatial-semantic relationships. However, the mIoU of LBYL-NET is somewhat disappointing. The problem lies in the landmark-aware convolution mechanism. Although it can establish relationships between images and text, this region-based convolution struggles when interpreting RS images with varying scales and complex content.

CNN-based methods often rely on manually designed visual–textual relation modeling, which limits their generalization and stability. Transformer-based methods can effectively solve this problem, as the attention mechanism in transformers is well-suited for interacting with and fusing image and text features, ultimately leading to accurate target localization. As illustrated in Table I, TransVG, VLTVG, MGVLF, and LPVA consistently outperform CNN-based approaches in most cases. Especially under the higher precision thresholds (e.g., Pr @0.9), the performance enhancements achieved by the transformer-based models are distinct. For example, compared with LBYL-Net-LSTM (i.e., the best CNN-based approach in Pr @0.9), the improvements of TransVG, VLTVG, MGVLF, and LPVA are 6.11%, 16.52%, 13.08%, and 13.56%, respectively. For mIoU, the advantages of transformer-based

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methods are also quite noticeable. For instance, TransVG, VLTVG, MGVLF, and LPVA are 0.0283, 0.0639, 0.0487, and 0.067 higher than ReSC, which is the best CNN-based approach in mIoU. The above positive results achieved by TransVG, VLTVG, MGVLF, and LPVA can be attributed to the transformers' superior capacity for modeling fine-grained cross-modal interactions through attention mechanisms.

Although the compared models produce good results, the proposed LPQR-DVG still outperforms them in all cases. Taking Pr @0.5 as an example, our LPQR-DVG achieves 78.69%, outperforming LPVA (the second-best model) by 3.48%. Taking mIoU as another example, LPQR-DVG achieves 0.6844, marking an improvement of 0.034 over LPVA. The superior performance of our LPQR-DVG model can be attributed to two points. First, the integration of text guidance throughout the visual feature extraction process significantly enhances visual–textual understanding by continuously aligning visual features with linguistic context. Second, the novel RSVG-oriented reversed U-Net architecture enables hierarchical, multistage cross-modal interaction through its progressive upsampling and down-sampling pathways, which systematically refine visual-linguistic alignment at different spatial scales. An encouraging point is that there are significant performance gaps between LPQR-DVG and other methods. This is due to the incorporation of a diffusion paradigm with our RSVG-oriented reversed U-Net, which revolutionizes the localization process by formulating it as an iterative denoising procedure. This approach enables the progressive refinement of noisy queries, leading to more accurate and robust identification of semantically relevant regions.

2) Analysis of Results on OPT-RSVG: As detailed in Table II, similar to the results on DIOR-RSVG, the performance of different methods based on OPT-RSVG is appropriate. For the CNN-based methods, LBYL-Net-LSTM and ReSC outperform others. Their Pr @0.5 values are as high as 72.82% and 73.24%. However, it is worth noting that LBYL-NET-BERT struggles to produce stable results on OPT-RSVG. These observations suggest that CNN-based methods, which often rely on manually designed visual–textual relation modeling, exhibit limitations in both stability and accuracy due to their information mining manner.

Transformer-based methods leverage attention mechanisms to effectively model interactions between image and text features, ultimately leading to more accurate target localization. Consequently, on the OPT-RSVG dataset, these methods consistently outperform CNN-based approaches in most cases, especially under stricter localization thresholds such as Pr @0.9. For example, compared to ReSC (i.e., the best CNN-based method in Pr @0.9), TransVG, VLTVG, MGVLF, and LPVA achieve improvements of 4.73%, 9.95%, 8.3%, and 6.04%, respectively. A similar advantage is observed in mIoU. For example, TransVG, VLTVG, MGVLF, and LPVA exceed ReSC (i.e., the best CNN-based model in mIoU) by 0.0211, 0.0364, 0.0311, and 0.0444, respectively.

Aligning with the observations on DIOR-RSVG, the proposed LPQR-DVG consistently outperforms all compared methods across all evaluation metrics. Taking Pr @0.5 as an example, our LPQR-DVG achieves 78.26%, outperforming LPVA (the second-best model) by 1.83%. Taking mIoU as another example, LPQR-DVG achieves 0.6693, marking an improvement of 0.0237 over LPVA. Collectively, the sustained high performance across both DIOR-RSVG and OPT-RSVG datasets underscores not only the robustness of our method but also its superior generalization capabilities to novel contexts.

Besides the numerical results, visual location maps achieved by different models are also provided and discussed. Due to space limitations, the detailed visual results and corresponding discussion can be found in the supplementary material.

TABLE III
ABLATION STUDY OF MAIN PARTS OF LPQR-DVG

Diffusion paradigm	Multiscale text guidance	PQR unit	DIOR-RSVG		OPT-RSVG	
			$Pr@0.5$ (%)	$mIoU$	$Pr@0.5$ (%)	$mIoU$
✓	×	×	68.59	0.5975	69.11	0.5955
✓	✓	×	<u>77.95</u>	<u>0.6774</u>	<u>77.07</u>	<u>0.6596</u>
✓	×	✓	72.89	0.6369	73.83	0.6356
×	✓	✓	76.46	0.6637	76.26	0.6401
✓	✓	✓	**78.69**	**0.6844**	**78.26**	**0.6693**

D. Ablation Studies

As introduced in Section III, the main parts of our LPQR-DVG are MTFLM, IQLM, and the diffusion training algorithm. Meanwhile, the key components in MTFLM and IQLM are the multiscale text guidance and the PQR unit. To investigate their contributions to LPQR-DVG, this section outlines the design of the ablation experiments.

First, we fix the diffusion training algorithm and examine the importance of multiscale text guidance and the PQR unit. To this end, the multiscale text guidance would be removed from LPQR-DVG, and the PQR unit would also be replaced with a standard cross-attention transformer unit. Note that only one of the two steps above is executed at a time. The results of different LPQR-DVGs counted on the two datasets are summarized in Table III. For clarity, we report their Pr @0.5 and mIoU values exclusively. When the multiscale text guidance and PQR unit are ablated (see the first row), LPQR-DVG's performance drops significantly, achieving only 68.59% Pr @0.5 and 0.5975 mIoU on DIOR-RSVG, and 69.11% Pr @0.5 and 0.5955 mIoU on OPT-RSVG. Once the multiscale text guidance is added (see the second row), the model's Pr @0.5 and mIoU values are enhanced by 9.36% and 0.0799 on DIOR-RSVG, and 7.96% and 0.0641 on OPT-RSVG. This means that incorporating multiscale text guidance enables early visual–text interaction during feature extraction, which reduces the complexity of later-stage vision-text alignment and enhances localization accuracy. Alternatively, when only the PQR unit is added (see the third row), LPQR-DVG achieves 72.89% and 0.6369 for Pr @0.5 and mIoU on DIOR-RSVG, and 73.83% and 0.6356 on OPT-RSVG. This indicates that the PQR unit in IQLM effectively aligns visual and textual features, enabling efficient interaction between queries, visual elements, and textual features, ultimately enhancing localization accuracy. Finally, when the multiscale text guidance and PQR unit are both incorporated (see the fourth row), the complete LPQR-DVG's behavior becomes the strongest. This implies that the combination of multiscale text guidance

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TABLE IV
ABLATION STUDY OF INTERACTION STAGES IN MTFLM

Interaction Methods (Backbone Stage)	#Params.	DIOR-RSVG		OPT-RSVG	
		$Pr@0.5$ (%)	$mIoU$	$Pr@0.5$ (%)	$mIoU$
{4}	197.65 M	77.56	0.6734	77.43	0.6632
{3,4}	202.40 M	78.13	0.6785	77.57	0.6652
{2,3,4}	203.78 M	78.15	0.6823	78.08	0.6683
{1,2,3,4}	204.22 M	78.49	0.6838	77.60	0.6645
{0,1,2,3,4}	204.22 M	**78.69**	**0.6844**	**78.26**	**0.6693**

TABLE V
ANALYSIS OF THE INFLUENCE OF LANGUAGE MODELS ON LPQR-DVG

Text Backbone	DIOR-RSVG		OPT-RSVG	
	$Pr@0.5$ (%)	$mIoU$	$Pr@0.5$ (%)	$mIoU$
BERT [45]	**78.69**	**0.6844**	**78.26**	**0.6693**
ALBERT [55]	71.84	0.6157	72.29	0.6183
RoBERTa [56]	77.71	0.6762	78.25	0.6687
CLIP [57]	70.31	0.6094	69.42	0.5963
RemoteCLIP [58]	69.92	0.6034	72.24	0.6170

and the PQR unit enables comprehensive vision-language alignment, facilitating effective cross-modal feature interaction and query-based feature learning, which leads to precise target localization.

Second, the diffusion training algorithm is changed to the standard training algorithm to determine if the diffusion paradigm is necessary or not. The results are also exhibited in Table III. By analyzing the scores in the fourth and fifth rows, we can easily see that LPQR-DVG with diffusion outperforms LPQR-DVG with the standard training strategy. Specifically, removing the diffusion algorithm results in declines of 2.23% and 0.0207 in Pr @0.5 and mIoU on DIOR-RSVG, and 2% and 0.0292 in Pr @0.5 and mIoU on OPT-RSVG. This denotes that the diffusion learning paradigm can progressively align visual–text features through its iterative refinement process, enabling gradual convergence to accurate target locations.

E. Further Analysis of MTFLM

Based on the ablation study results (see Section IV-D), the usefulness of multiscale text guidance in MTFLM is confirmed. This section will investigate its significance to our LPQR-DVG from two perspectives: the role of text guidance in visual feature extraction, and the impact of different text encoders on the final results.

1) Text Guidance in Visual Feature Extraction: As discussed in Section III-B, the textual clues are embedded into all stages of ResNet-50. Here, the multiscale text guidance would be analyzed by varying the incorporation stages from deep to shallow. The results of various LPQR-DVGs are exhibited in Table IV. For clarity, we only select Pr @0.5 and mIoU as the assessment criteria. Meanwhile, the LPQR-DVGs' parameter volume (#Params.) is also chosen for the evaluation. By observing the table, we can see that as the number of incorporating stages increases, LPQR-DVGs' performance improves with an acceptable parameter volume growth. Taking results on DIOR-RSVG as an example, when implementing text guidance at stage 4, LPQR-DVG achieves 77.56% on Pr @0.5 and 0.6734 on mIoU, and its parameter volume is equal to 197.65 M. When implementing text guidance at all stages, the model achieves 1.13% and 0.011 improvements in Pr @0.5 and mIoU, respectively, and its parameter volume increases by only 6.57 M. Similarly, on OPT-RSVG, implementing text guidance at all stages can also yield improvements of 0.83% in Pr @0.5 and 0.0061 in mIoU compared to implementing text guidance only at Stage 4. This indicates that implementing text guidance across all network stages enables comprehensive visual–text interaction, achieving cross-modal correspondences from low-level textures to high-level semantics. Therefore, our LPQR-DVG model adopts guidance from all five stages as the default setting. In addition to the numerical results, we still investigate the attention maps of each stage to analyze their distinct contributions to textual information processing. The details can be found in the supplementary material.

2) Different Text Encoders in MTFLM: As detailed in Section III-B, MTFLM adopts BERT [45] as the text backbone of its textual feature extractor. To investigate whether this choice is suitable, we replace BERT with several other popular language models, including ALBERT [55], RoBERTa [56], CLIP [57], and RemoteCLIP [58]. ALBERT and RoBERTa focus on general language understanding, while CLIP and RemoteCLIP are specialized for visual–text alignment tasks. The Pr 0.5 and mIoU scores of LPQR-DVGs with different language models counted on DIOR-RSVG and OPT-RSVG are summarized in Table V.

As detailed in the table, the behavior of LPQR-DVGs with different textual backbones is acceptable. Among the compared textual backbones, CLIP and RemoteCLIP demonstrate weaker performance. For DIOR-RSVG, CLIP and RemoteCLIP achieve 70.31% and 69.92% on Pr 0.5, with 0.6094 and 0.6034 on mIoU, respectively. For OPT-RSVG, their Pr 0.5 scores are 69.42% and 72.24% on Pr 0.5, and their mIoU values are 0.5963 and 0.6170. This is because CLIP and RemoteCLIP pay more attention to building the relationships between RS images and scene-level semantics. Their capacity to interpret complex textual descriptions is relatively low, which limits their effectiveness in RSVG tasks. The strongest one is RoBERTa, with Pr 0.5 and mIoU scores of 77.71% and 0.6762, and 78.25% and 0.6687 on DIOR-RSVG and OPT-RSVG, respectively. Compared to them, LPQR-DVG with BERT achieves the best performance. For Pr 0.5, its improvements are 6.85% and 5.97% (over ALBERT), 0.98% and 0.01% (over RoBERTa), 8.38% and 8.84% (over CLIP), and 8.77% and 6.02% (over RemoteCLIP) on DIOR-RSVG and OPT-RSVG. For mIoU, its enhancements are 0.0687 and 0.051 (over ALBERT), 0.0082 and 0.0006 (over RoBERTa), 0.075 and 0.073 (over CLIP), and 0.081 and 0.0523 (over RemoteCLIP) on the two datasets. These findings illustrate that choosing BERT as the textual backbone is suitable for our tasks.

F. Further Analysis of IQLM

The usefulness of the PQR unit in IQLM has been confirmed in Section IV-D. This section will further

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TABLE VI
ANALYSIS OF U-NET RESIDUAL PART IN IQLM

U-Net Residual	Query Refinement	DIOR-RSVG $Pr@0.5$ (%)	DIOR-RSVG $mIoU$	OPT-RSVG $Pr@0.5$ (%)	OPT-RSVG $mIoU$
×	✓	77.61	0.6734	77.46	0.6612
✓	✓	**78.69**	**0.6844**	**78.26**	**0.6693**

TABLE VII
ANALYSIS OF RSVG-ORIENTED REVERSED U-NET CHANNEL SETTINGS

Reversed U-Net Init. Channel Number	#Params.	DIOR-RSVG $Pr@0.5$ (%)	DIOR-RSVG $mIoU$	OPT-RSVG $Pr@0.5$ (%)	OPT-RSVG $mIoU$
8	201.85 M	77.57	0.6761	77.20	0.6602
32	204.22 M	**78.69**	**0.6844**	**78.26**	**0.6693**
64	210.95 M	78.24	0.6787	78.25	0.6684
128	236.59 M	78.56	0.6828	76.77	0.6565
256	336.68 M	78.61	0.6797	77.80	0.6664

investigate its significance to our LPQR-DVG. As explained in Section III-C, the PQR unit consists of a U-Net residual part and a query refinement part. The U-Net residual part aims to capture hierarchical feature representations from the input data, while the query refinement part focuses on refining these representations by leveraging interactions between visual and textual features, ultimately achieving accurate target localization. The two parts of the PQR unit perform distinct roles, providing complementary information to support the final location. To further study their contributions to LPQR-DVG, we conduct the following ablation experiments. First, the U-Net residual part is removed from the PQR unit. In other words, the query refinement part works alone. Second, we call back the U-Net residual part to construct a complete PQR unit.

The Pr @0.5 and mIoU scores of LPQR-DVGs with only the query refinement part and the complete PQR unit are exhibited in Table VI. It is obvious that the complete PQR unit improves the overall performance of LPQR-DVG. Taking results on DIOR-RSVG as an example, when the U-Net residual part is removed (see the first row), LPQR-DVG's Pr @0.5 and mIoU values are as low as 77.61% and 0.6734. Once the U-Net residual part is added to the PQR unit (see the second row), the model's Pr @0.5 and mIoU values are enhanced by 1.08% and 0.011. Similar results can be observed on OPT-RSVG, where incorporating the U-Net residual component into the PQR unit enhances the model's Pr @0.5 and mIoU by 0.8% and 0.0081, respectively. These findings suggest that the inclusion of the U-Net residual part enhances feature representation optimization, thereby improving localization accuracy. In addition, although the behavior of LPQR-DVG with only the query refinement part is weaker, its scores are still fair. This implies that the query refinement part's potential for independent work.

G. Free Settings Analysis

1) Channel Settings for RSVG-Oriented Reversed U-Net: In LPQR-DVG, the RSVG-oriented reversed U-Net plays the crucial role in obtaining predicted coordinates. To explore the impact of different channel settings on the RSVG-oriented reversed U-Net's performance, we change the channel numbers from 8 to 256. In these cases, LPQR-DVGs' Pr @0.5 and mIoU counted on the two datasets are summarized in Table VII. We can observe that as the channel number increases, the behavior of LPQR-DVG first strengthens and then weakens, and the peak values of Pr @0.5 and mIoU appear when the channel number equals 32. In addition, the model sizes grow with the number of channels. These examinations suggest that when the initial channel number is too small, the model suffers from inadequate feature representation capacity, resulting in compromised semantic understanding. Meanwhile, the large channel numbers would introduce unnecessary complexity, leading to overfitting and a degradation in localization accuracy. In summary, the number of channels should be carefully selected within an appropriate range. As a result, in this article, we set the initial channel number to 32.

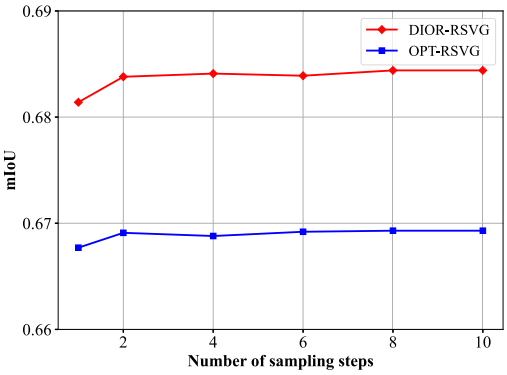


Fig. 7. Analysis of diffusion inference sampling steps. As the number of inference steps increases, the performance becomes stable.

2) Inference Sampling Steps: In the inference phase, the DDPM [51] strategy is employed for the final prediction. Thus, different sampling steps may lead to varying results during inference. To ensure a suitable number of sampling steps for our LPQR-DVG, we vary it from 1 to 10 and count the mIoU values of LPQR-DVGs counted on the two datasets for clarity. The curves are drawn in Fig. 7. When the sampling steps are limited (e.g., the number of steps is fewer than 6), the model's performance fluctuates distinctly. For example, with only one step, LPQR-DVG achieves merely 0.6814 mIoU on DIOR-RSVG, as insufficient steps prevent the full utilization of the diffusion paradigm's iterative refinement capability. When increasing to two and four steps, the model achieves improvements of 0.0024 and 0.0027 on mIoU, demonstrating a gradual enhancement through multistage denoising. The performance eventually stabilizes when exceeding six sampling steps, reaching a plateau of 0.6844 mIoU with ten sampling steps, indicating convergence of the diffusion process's error correction capacity. Similar results can be observed on OPT-RSVG. LPQR-DVG achieves 0.6677 mIoU on the OPT-RSVG dataset with only one sampling step. When the number of sampling steps is increased to 10, the model achieves a stable mIoU of 0.6693. As a result, in this article, the inference step is set to ten as the default. Following this, to further investigate the distinct roles of each sampling step, we

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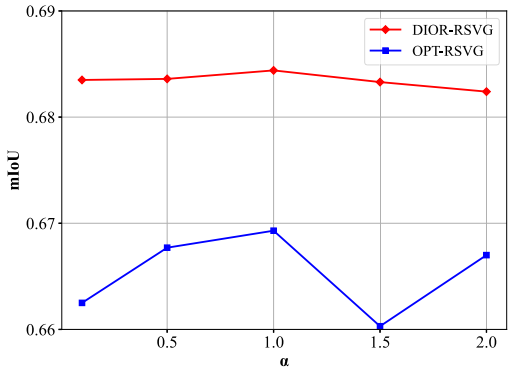


Fig. 8. Influence of α on LPQR-DVG. The mIoU curves are counted on DIOR-RSVG and OPT-RSVG.

conduct visualizations of the target localization results at each sampling step for the ten inference steps setting. Due to space limitations, the visual results and corresponding discussion can be found in the supplementary material.

3) Loss Weight Parameters of LPQR-DVG: Our loss function $\mathcal{L}_{\text{train}}$ [see (7)] consists of two parts, i.e., the box localization loss $\mathcal{L}_{\text{train}}^{\text{box}}$ and auxiliary box refinement loss $\mathcal{L}_{\text{train}}^{\text{aux}}$. $\mathcal{L}_{\text{train}}^{\text{box}}$ is designed to jointly optimize precise coordinate regression and geometric consistency in the diffusion process. $\mathcal{L}_{\text{train}}^{\text{aux}}$ is developed to enhance feature learning through deep supervision. To deeply investigate their functions, we tune the values of the hyperparameter α in a range of $\{0.1, 0.5, 1, 1.5, 2\}$. The mIoU scores of LPQR-DVGs with different loss functions counted on the two datasets are displayed in Fig. 8. By observing the curves, we can find the following points. Taking the DIOR-RSVG dataset as an example, when α is set to 1, LPQR-DVG achieves the optimal mIoU of 0.6844. As α continues to increase, the mIoU exhibits a monotonic decrease, eventually reaching 0.6824 at $\alpha = 2$, which corresponds to a reduction of 0.0020 compared to the performance at $\alpha = 1$. Moreover, when smaller values of α are used (e.g., $\alpha = 0.5$), mIoU declines by 0.0008, and the training process exhibits noticeably reduced stability under these conditions. Similar results can be observed on the OPT-RSVG dataset. When $\alpha = 1$, LPQR-DVG's mIoU reaches its maximum value of 0.6693. When $\alpha = 0.5, 0.1$ and $\alpha = 1.5, 2$, the mIoU decreases by 0.0016, 0.0068, 0.009, and 0.0024, respectively. This trend confirms that both large and small values of α adversely affect localization performance. Based on the above discussion, we suggest that the optimal values of α for different tasks can be fixed around 1.

H. Parameter Volume and Inference Speed of LPQR-DVG

This section analyzes the parameter volume and inference time costs of our LPQR-DVG. The comparisons between LPQR-DVG and other methods are summarized in Table VIII. Our model incurs moderately higher parameter sizes and computational costs than efficient alternatives, such as LPVA and VLTVG. The reasons behind these findings are twofold. One is the complexity of the model, and the other is the number of diffusion sampling steps.

Regarding model parameters, LPQR-DVG employs an input size of 800×800 and facilitates multistage progressive interaction through the MTFLM and IQLM modules. In contrast, the compared methods utilize a smaller input size of 640×640. Consequently, LPQR-DVG exhibits a slight increase in model parameters, with 75.82 M over LPVA and 59.19 M over VLTVG. However, this parameter overhead is acceptable given the significant improvement in the model's overall performance across the DIOR-RSVG and OPT-RSVG datasets. For the inference speed, LPQR-DVG spends more time compared to other methods, as it is significantly affected by the diffusion process. As shown in Table VIII, the inference time increases with additional inference steps. Fortunately, all of them remain within the millisecond range. Considering that the diffusion process can enhance both the precision and prediction stability of LPQR-DVG, the current moderate computational overhead is worthwhile. Speeding up the inference process is one of our future works.

TABLE VIII
PARAMETER VOLUME AND INFERENCE SPEED OF LPQR-DVG

Model Name	#Params.	Inference Speed (ms)
CNN-based:		
FAOA (BERT) [28]	174.40 M	30
FAOA (LSTM) [28]	74.44 M	17
ReSC [30]	171.52 M	30
LBYL-NET (BSRT) [29]	169.40 M	28
LBYL-NET (LSTM) [29]	69.38 M	18
transformer-based:		
TransVG [33]	142.81 M	29
VLTVG [35]	145.03 M	32
MGVLF [2]	145.61 M	37
LPVA [1]	128.4 M	30
LPQR-DVG (step=1)	204.22 M	69
LPQR-DVG (step=2)	204.22 M	126
LPQR-DVG (step=5)	204.22 M	289
LPQR-DVG (step=10)	204.22 M	621

V. CONCLUSION

This article proposes a new model for RSVG tasks named LPQR-DVG. It primarily consists of an MTFLM module and an IQLM module. MTFLM is designed to continuously guide visual features with textual information throughout the feature extraction process, ensuring more robust and contextually aligned interactions between visual and text features. IQLM is developed to enable sufficient interaction between queries, visual, and text features during the training process in the RSVG-oriented reversed U-Net, enhancing the model's understanding capabilities of discriminative details in both text and visual features. To train LPQR-DVG properly, we introduce a progressive diffusion learning strategy. By progressively denoising noisy bounding box coordinates through iterative interactions with visual and textual features in the RSVG-oriented reversed U-Net, more accurate localization results can be achieved. Positive experimental results based on the DIOR-RSVG and OPT-RSVG datasets indicate that our model shows promising results for the RSVG task.

Although our model demonstrates competitive performance, there remains room for improvement. First, the visual features used in IQLM, while capturing both local and global information, are limited to a single scale. It would be beneficial to explore the use of multiscale visual features from MTFLM to

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enhance interaction with textual features. Second, compared with existing methods, LPQR-DVG still shows limitations in terms of model size and inference speed. Therefore, further improving the model's performance while reducing its size and enhancing inference speed will be the focus of our future research.

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